Pick the rows with WHERE

In Ep 04 you picked columns. In Ep 05 you pick rows: only the orders you want. Here is every line from the video, with what each part means, so you can type along. You need the **orders** table from Ep 03 (ep05_where.sql makes it if you missed it).

SELECT
Picks the columns. The ones going down.

WHERE
Picks the rows. The ones going across. Only rows that match your rule come back.

'single quotes'
A word goes inside single quotes: `'Leeds'`

Numbers
No quotes: `10`

Before step 1 `look at the whole table first`

```
USE shop;
SELECT * FROM orders;
```

Open Workbench, click your connection, and run both lines. You see all 3 orders. Now we only want one of them.

order_id	city	amount
1001	London	12.99
1002	Leeds	4.50
1003	Hatfield	21.50

Step 1 · WHERE with text `words go in single quotes`

```
1  SELECT * FROM orders WHERE city = 'Leeds';
```

I only want the orders from Leeds. Leeds is a word, so it goes inside single quotes. Run it, and only one row comes back:

order_id	city	amount
1002	Leeds	4.50

city	is it Leeds?
~~London~~	no, throw it out
Leeds	yes, keep it
~~Hatfield~~	no, throw it out

So what did SQL do? It checked every row, one by one, and asked: is the city Leeds? If yes, it keeps the row. If not, it throws it out. That's all WHERE does.

Capital letters: in MySQL, `'Leeds'` and `'leeds'` find the same row. If your table shows leeds in small letters, that's fine. Added order 1004 in Ep 03's your turn? Then you may see one more row on this sheet. That's fine too.

Step 2 · WHERE with numbers no quotes this time

```
1  SELECT * FROM orders WHERE amount > 10;
```

I want the orders above 10 pounds. It works the same way, but numbers don't need quotes. Run it, and two rows come back. Leeds is gone, because 4.50 is not more than 10.

order_id	city	amount
1001	London	12.99
1003	Hatfield	21.50

>	more than
<	less than
=	equal to

Forgot which sign it is? Remember the crocodile from school. Its mouth always opens to the bigger side. In `amount > 10` the mouth opens to amount, so amount has to be the bigger one.

Break it on purpose

```
1  SELECT * FROM orders WHERE city = leeds;
```

Take the quotes off, and you get an error. Without quotes, SQL thinks leeds is a column name, like city or amount. And we don't have a column called leeds. Put the quotes back and it works.

`✗ Error Code: 1054. Unknown column 'leeds' in 'where clause'`

So remember: words go inside quotes, numbers don't. `city = 'Leeds'` has quotes. `amount > 10` has none.

No error, but no rows? 0 row(s) returned

Check the spelling inside the quotes

`WHERE city = 'Leds'` gives you an empty table and no error. SQL looked for Leds, and no row has it. Fix the spelling and run it again.

Your turn

Find the orders that are **below 10 pounds**. Then comment the **city** you get on the reel.

Next: **AND, OR, IN**. How to use two rules at once.

Answer: SELECT * FROM orders WHERE amount < 10; gives one row, order 1002, so the city is Leeds.